Predicting Heart Disease from Survey Data: Linear Models versus Boosted Trees in NumPy

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***Abstract*—We predict if a person has coronary heart disease from 321 answers of the 2015 BRFSS survey, using only NumPy. The linear models from the course underfit and stay below a validation F1 of 0.42. One-hot encoding improves Ridge regression from 0.416 to 0.427, and our own gradient-boosted decision trees reach 0.452 on validation and 0.454 on the AIcrowd test set. We also show that the decision threshold matters a lot, and that one survey question gives away the label for 2.7% of the positive cases.**

I. Introduction

For every person we predict if they have coronary heart disease (+1) or not (-1). We have 328,135 labelled and 109,379 test persons with 321 features: numbers, coded answers (e.g. 7 = "don't know") and many missing values (147 features are missing for more than half of the persons). Only 8.83% of the persons are positive, so we use the F1 score of the positive class, $F_1 = 2PR/(P + R)$ with precision P and recall R. Accuracy is not useful here: always answering -1 gives 91.2% accuracy but $F_1 = 0$.

II. How we evaluate

We split the labelled data per class into 80% training and 20% validation (seed 42): 262,508 and 65,627 rows (5,795 positives). All models use this split, and everything the preprocessing learns (categories, medians, means, standard deviations, bin edges) comes from the training rows only. Each model gives a score, and we predict +1 when the score is at least a threshold t, chosen to maximize the validation F1 (grid with step 0.01). Resampling the validation rows changes F1 by about 0.005, so we treat smaller differences as ties. Because we also choose the hyperparameters on these rows, our validation F1 is a bit optimistic. We used one hold-out split instead of k-fold cross-validation because one tree model needs 5 to 15 minutes to train. As an independent check, we retrain the final models on all labelled rows with the same settings and submit them to AIcrowd.

III. Feature processing

Reviewed rules. We checked some features in the CDC codebook [4]. We remove `SEQNO` and `_PSU` (the same record number), set "don't know" and "refused" of `GENHLTH` (7, 9) and `PHYSHLTH` (77, 99) to missing, set `PHYSHLTH` 88 ("none") to 0 days, and add a missing indicator for both features.

Missing values carry information. Many answers are missing on purpose, because some questions are only asked to some persons. For example, `BPMEDS` (blood pressure medication) is only answered by the 40% of persons with high blood pressure: 16.3% of them are positive, but only 3.8% of the others.

Linear models. We fill missing values with the training median. In the one-hot version, each column with at most 10 different training values (223 columns) is replaced by one 0/1 column per value seen in training plus one for missing values (a new value gives zeros). Then we standardize all columns and add a bias column (322 columns, or 1,283 with one-hot). **Trees** use the raw values put into bins: one bin per value if a feature has fewer than 64 values, otherwise 63 quantile bins, and one bin for missing values.

Label leakage. `HAREHAB1` asks about rehabilitation after a heart attack, so only persons who reported one (which is part of the label definition) answer it: all 773 of them are positive (2.7% of the positives). It is also in the test data, so we kept it, but we report it here (`STREHAB1`, after a stroke, is a weaker case with 36% positives).

IV. Models

Required methods (`implementations.py`). We implemented linear regression with gradient descent and with SGD (one sample per step) on $\mathcal{L} = \frac{1}{2N}\|y - Xw\|^2$, least squares and Ridge regression with the normal equations $(X^\top X + 2N\lambda I)w = X^\top y$, logistic regression with gradient descent on $\frac{1}{N}\sum_n [\log(1+e^{x_n^\top w}) - y_n x_n^\top w]$ (with `logaddexp` for stability), and its regularized version with $\lambda\|w\|^2$ (gradient term $2\lambda w$). All six pass the public grading tests of the course. In our experiments we use logistic regression ($\gamma = 0.01$, 5,000 steps, $w_0 = 0$, labels 0/1) and Ridge regression on the labels ± 1 with $\lambda \in \{10^{-6}, 10^{-5}, \ldots, 1\}$; with a small λ, Ridge gives the least-squares solution that the two MSE methods converge to. We did not tune the regularized logistic regression, because the linear models already underfit and a larger λ only made Ridge worse.

Gradient-boosted trees (`boosting.py`). Because the linear models underfit, we implemented a non-linear model, following XGBoost [1], [2]. A tree asks a few yes/no questions about the features, and boosting adds many small trees, each one correcting the errors of the previous ones. We start from the log-odds of the training positive rate, and each tree

Table I
VALIDATION F1 (BEST THRESHOLD) AND AICROWD TEST F1.

Model	Val. F1	Test F1
Always -1 (accuracy 0.912)	0.000	–
Logistic regression, GD ($t = 0.19$)	0.418	–
Ridge, $\lambda = 10^{-5}$ ($t = -0.58$)	0.416	–
Ridge + one-hot, $\lambda = 10^{-5}$ ($t = -0.60$)	0.427	0.436
Trees $d = 1$, $\eta = 0.1$, 400 trees	0.444	–
Trees $d = 3$, $\eta = 0.1$, 331 trees ($t = 0.21$)	0.449	0.452
Trees $d = 5$, $\eta = 0.1$, 224 trees	0.444	–
Trees $d = 3$, $\eta = 0.05$, 800 trees ($t = 0.21$)	**0.452**	**0.454**

Table II
ABLATIONS (VALIDATION F1).

Ridge: feature processing	
Original features	0.414
+ reviewed rules	0.416
+ one-hot (without rules)	0.428
+ reviewed rules + one-hot	0.427
Ridge: regularization (reviewed)	
$\lambda = 10^{-5}$ / 10^{-2} / 1	0.416 / 0.412 / 0.394
Trees: tuning	
$d = 3$, $\eta = 0.1$ (331 trees)	0.449
$d = 4$, $\eta = 0.1$ (234 trees)	0.446
$d = 3$, $\eta = 0.05$ (798 trees)	0.452
$d = 4$, $\eta = 0.05$ (488 trees)	0.450
Average of 3 seeds ($d = 4$, $\eta = 0.05$)	0.450
Blend, Ridge weight 0 / 0.2 / 0.5	0.451 / 0.448 / 0.442

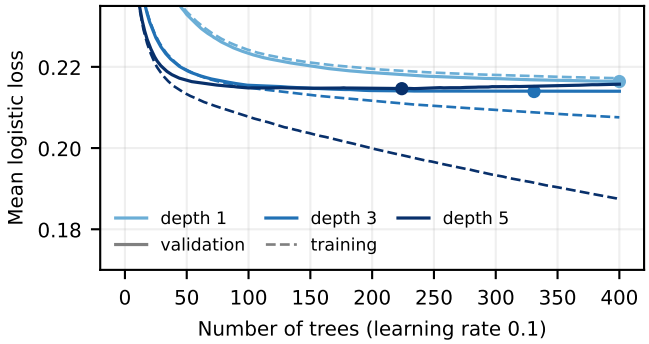


Figure 1. Training (dashed) and validation (solid) logistic loss of the boosted trees. Dots: lowest validation loss. Depth 1 underfits (both curves together), depth 5 overfits (the validation loss goes up after 224 trees).

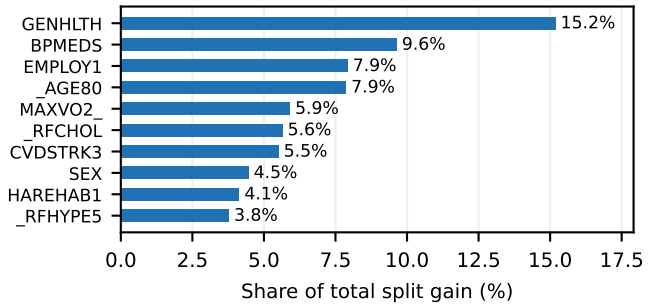


Figure 2. Top 10 features of the $d = 3$, $\eta = 0.1$ trees by share of the total split gain.

is fitted to the gradient $g = p - y$ and Hessian $h = p(1-p)$ of the logistic loss. From the sums of g and h per node, feature and bin we get, for every threshold (and both sides for missing values), the gain $\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}$, and we keep the best split. A leaf predicts the Newton step $-G/(H+\lambda)$ with $\lambda = 1$, and each child needs $H \geq 1$. Trees grow level by level up to depth d, each uses a random half of the training rows [3] and is added with learning rate η. We keep the number of trees with the lowest validation logistic loss. We tested our split search against a brute-force search.

V. RESULTS

Table I compares the models and Table II shows the effect of each change. Our final model (last row of Table I) has a validation precision of 0.376, recall of 0.567 and accuracy of 0.879.

The linear models underfit. The training and validation losses are almost the same (logistic: 0.2282 and 0.2284; Ridge: 0.1358 and 0.1360), and F1 only goes down when λ grows.

One-hot encoding improves F1 by 0.011 to 0.014, because codes like "don't know" and missing values get their own weight instead of being used as numbers. With one-hot, our hand-written rules do not matter anymore.

The trees improve F1 by 0.024 over Ridge with one-hot on validation and by 0.018 on the test set. Depth 1 trees (one split each, so no interactions) already give most of this gain, so the main benefit comes from thresholds on single features and the missing bins. Depth 3 adds a little (validation loss from 0.2164 to 0.2139), so some interactions help, while depth 5 overfits (Fig. 1). A smaller learning rate gives a small gain. Averaging seeds or blending with Ridge does not help more than the noise, so the trees already contain what Ridge learns.

The threshold matters a lot. With 8.8% positives, few probabilities are above 0.5. For the $d = 3$, $\eta = 0.1$ trees, F1 is 0.251 at $t = 0.5$ (precision 0.61, recall 0.16) but 0.449 at $t = 0.21$.

Important features. The trees mostly use general health, blood pressure medication, employment (retired or unable to work), age, cholesterol, stroke and sex (Fig. 2), all known risk factors. The leaking `HAREHAB1` gives 4.1% of the gain.

VI. CONCLUSION

The linear models underfit these data. One-hot features help, but our boosted trees help more (test F1 0.454), and the validation and test scores agree (Table I). `run.py` reproduces our best submission in about 15 minutes (fixed seeds), and `train.py` and `tune_boosting.py` reproduce both tables.

REFERENCES

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